Practical Activity: Spark SQL

Exercise 1:

A city wants to analyze the usage of its public bike-sharing system. You are given a dataset of bike rental transactions. Your task is to use **Spark SQL** to extract insights about user behavior, station popularity, peak hours, and overall system performance.

Dataset (bike_sharing.csv)

The structure of Dataset is as follow:

rental_id,user_id,age,gender,start_time,end_time,start_station,end_station,duration_minutes,price

Column Descriptions

- **rental_id** – unique ID for each rental
- **user_id** – unique ID of the user
- **age** – user's age
- **gender** – M or F
- **start_time** – rental start timestamp
- **end_time** – rental end timestamp
- **start_station** – where the bike was picked up
- **end_station** – where the bike was returned
- **duration_minutes** – length of the trip
- **price** – rental cost in dollars

1. Data Loading & Exploration

1. Load the CSV file into a Spark DataFrame.

2. Display the schema.
3. Show the first 5 rows.
4. How many rentals are in the dataset?

2. Create a Temporary View

Create the SQL view: bike_rentals_view

3. Basic SQL Queries

1. List all rentals longer than **30 minutes**.
2. Show all rentals starting at **"Station A"**.
3. Calculate the **total revenue** (sum of the column price).

4. Aggregation Queries

1. Count how many rentals were made **from each start station**.
2. Compute the **average rental duration** per start station.
3. Identify the station with the **highest number of rentals**.

5. Time-Based Analysis

1. Extract the **hour** from start_time.
2. Count how many bikes were rented **per hour** (identify peak hours).
3. Determine the **most popular start station** during the morning (7–12).

6. User Behavior Analysis

1. Compute the **average age** of users.
2. Count users by **gender**.
3. Find which **age group** rents bicycles the most:
    - 18–30
    - 31–40

- 41–50
- 51+